Assignment 1

Explore following 4 websites and answer the questions

- **bitcoin.org**
- Blockchain.com
- https://coinmarketcap.com/
- https://www.investopedia.com/

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom? Satoshi Nakamoto

2. Bitcoin transactions are recorded on which data structure? Ledger

3. Bitcoin uses which consensus mechanism? Proof of Work

4. The smallest unit of Bitcoin is called? Satoshi

5. Bitcoin is a **peer-to-peer** type of what system? Network system

Wallets

6. A wallet stores the user's _private_ key.

7. The recovery phrase used in wallets is called a ___seed___ phrase.

8. Hardware wallets store keys __offline____ (online/offline).

9. Bitcoin addresses are derived from ___public___ keys.

10. What is required to sign a transaction? (public/private) private key

Security

11. Bitcoin prevents double-spending using ___blockchain___.
12. Bitcoin is secured using ___asymmetric___ cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin __core____.
14. The process of verifying transactions is called __mining____.
15. The Bitcoin network is described as ___decentralized___ (centralized/decentralized).

Mining

16. Miners solve ___cryptographic___ puzzles to add blocks.
17. Mining rewards a block with newly created __bitcoin____.
18. The block time target for Bitcoin is ___10___ minutes.
19. Mining difficulty adjusts every __2016____ blocks.
20. Miners compete to find a valid block __hash____ (hash/signature).

Blockchain Structure

21. Each block contains a ___merkle___ root.
22. Blocks are linked using the previous block's __hash____.
23. A Bitcoin transaction contains inputs and __hash____.
24. Unspent transaction outputs are called ___UTXOs___.
25. The first-ever Bitcoin block is called the __genesis____ block.

Network

26. A full node stores the entire blockchain __history____.

27. Bitcoin uses which protocol for communication? (TCP/P2P) P2P

28. A light wallet uses ___SPV___ mode (SPV/full).

29. Nodes relay information across a ___P2P___ network.

30. The maximum Bitcoin supply is **___21___** million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height: 960828
- Block hash: 00000000000000000000f9231726d7648d8a8bef0c0e0e4da85b87d18a331e12
- Timestamp: 03 Aug 2026 05:50:30 UTC
- Number of transactions: 25911
- Miner/pool name: AntPool

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction.
    184089545200cfb6ca35dfe87e7f231dbcf4e2487469485882d9acf56e13aa9b
11. Number of inputs: 5
12. Number of outputs: 4
13. Transaction fee: 4.7 sats/vB
14. How many confirmations does it have? 1

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.
16. Record: 1PuJjnF476W3zXfVYmJfGnouzFDAXakkL4

- Total BTC received: 47.159352 BTC
- Total BTC sent: 21.1265 BTC
- Current balance: 8.921578 BTC
  What is the role of a full node?

List any two developer resources available on bitcoin.org.

https://developer.bitcoin.org/devguide/block_chain.html

https://developer.bitcoin.org/devguide/mining.html

Explore : https://coinmarketcap.com/

Task C1: Market Information

- Current price of BTC: 62756$
- 24-hour % change: 1.32%
- Market capitalization: 1.25T$
- Circulating supply: 20.06M
- Total/maximum supply: 21M
- **Task C2: Historical Data**
- What was the price of Bitcoin one year ago? 113730 USD
- Note the highest and lowest prices in the last 30 days. 66160 USD, 61351 USD

Explore https://www.investopedia.com/

Why is Bitcoin considered volatile?

Bitcoin (BTC) is considered volatile due to a fixed supply cap, heavy market speculation, and an evolving financial market.Bitcoin (BTC) is considered volatile due to a fixed supply cap, heavy market speculation, and an evolving financial market.

List two investment risks mentioned.

- Loss of coins via theft, key mishandling
- Extreme volatility

- Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?
bitcoin.org gave the clearest understanding, as it holds the whitepaper and entire reasoning of why things are designed the way they are.

In one paragraph, explain how blockchain ensures trust without a central authority.
By using consensus models to split trust across nodes instead of relying on simply 1. The P2P model relies on “majority” of nodes agreeing still, but this requires more than 50% of the nodes to be compromised to be attacked successfully.